

Gym Workout Recommender

From lecture techniques to a practical workout plan

CHAPTER 4

Content-based

CHAPTER 8

Context-aware

CHAPTER 10

Weighted hybrid

Streamlit application • Local exercise catalogue • Explainable ranking

Match the workout to the user.

Training goals

General Fitness
Weight Loss
Muscle Gain
Strength

Practical inputs

Experience
Equipment and muscle focus
Available minutes
Location and current energy

Useful output

Ranked exercises
A time-budgeted plan
Reasons for each selection
Feedback for later sessions

Objective: personalized exercise selection with clear, inspectable ranking reasons.

Three methods. One clear workout builder.

Content-based

Matches exercise attributes to your goal and liked exercises.

Context-aware

Uses goal/experience rules and personal completion history.

Hybrid

Combines content, rating evidence, knowledge and context.

Goal and experience are separate choices.

General Fitness / Weight Loss / Muscle Gain / Strength • Beginner / Intermediate / Advanced

One catalogue, separate sources of evidence.

2,872

unique exercise entries

580

instruction matches

24

exercises with local photos

Exercise catalogue

Kaggle exercise descriptions, muscles, equipment, levels and ratings.
Detailed instructions are matched from Free Exercise DB.

Feedback and demonstrations

SQLite stores exercise ratings and completion.
48 local photos illustrate 24 matched exercises.
Missing images are clearly marked.

Where each part of the app comes from.

Lecture topic	Application use
Ch. 4 — Cosine similarity	Content-based; TF-IDF is our representation choice
Ch. 7 — Knowledge-based	Shared hard constraints and goal/experience rules
Ch. 8 — Context-aware	Workout context; completion-based score adaptation
Ch. 10 — Weighted hybrid	Weighted sum of component scores
Internal rating evidence	Ch. 3 popularity prior; no separate selectable mode

Same similarity principle, exercise features.

$$\text{Similarity}(q, x) = (q \cdot x) / (\|q\| \times \|x\|)$$

q = goal/focus vector; x = exercise vector

In the lecture

Combine movie attributes, build count vectors and compare them with cosine similarity.
Rank movies similar to a liked movie.

In our app

TF-IDF encodes exercise attributes. Match goal and muscle focus.
With ratings ≥ 4 : 60% profile similarity + 40% liked-exercise similarity.

Adapt the context concept to a workout.

In the lecture

Filter items by time or location availability.

The time-aware food example ranks by preference score $\times$ time relevance.

Context changes what is suitable.

In our app

Location restricts equipment; low energy limits difficulty.

The Context score combines knowledge rules with personal completion history.

Time budget controls plan allocation.

Conceptual adaptation: the app does not use the lecture's time-of-day scoring formula.

Apply feasibility rules across every method.

- | | | |
|-----------|---------------------------|--|
| 01 | Equipment | Required equipment must be available; bodyweight is always included. |
| 02 | Experience | Reject exercises above the experience limit; low energy caps it at beginner. |
| 03 | Muscle preferences | Respect focus and excluded primary muscles. |
| 04 | Diversity | Subtract 0.10 per already-selected exercise with the same primary muscle. |
| 05 | Time allocation | Reserve 5 minutes; allocate 5- or 7-minute exercise blocks as applicable. |

Chapter 10: add weighted score contributions.

Hybrid score = $\sum \text{weight} \times \text{component score}$

Each component contributes to a shared score; example weights sum to 1.

Two-component lecture example

Hybrid = 0.60 CF + 0.40 Content

For CF = 4.8 and Content = 4.2:
Hybrid = 2.88 + 1.68 = 4.56.

Generalized lecture example

Hybrid = 0.50 CF + 0.30 Content + 0.20 Context

The lecture also accepts a dictionary of score columns and weights.

The lecture mechanism, adapted to gym signals.



Content	Goal/focus and liked items
Popularity	Source prior and local ratings
Knowledge	Goal/category and experience
Context	Knowledge plus completion

$$B = 0.40C + 0.20P + 0.25K + 0.15X \cdot \text{Weights are project defaults.}$$

engine.py: Hybrid score; Chapter 10: generalized weighted sum

Inspect how each signal contributes.

Signal	Input	Weight	Contribution
Content C	0.60	0.40	0.24
Popularity P	0.70	0.20	0.14
Knowledge K	1.00	0.25	0.25
Context X	0.80	0.15	0.12

Base hybrid score

0.75

Illustrative inputs using the actual engine.py formula; these are not measured recommendations.

Feedback changes scores, not the selected method.

With no exercise history

Content uses the goal/focus query.
Popularity uses a source-rating prior.
Completion defaults to 0.5.

All three methods can produce a workout from the profile alone.

After logging activity

Ratings of 4 or 5 inform liked-exercise similarity.
Local ratings update rating evidence.
Completion updates Context.

Hybrid keeps the same 40/20/25/15 weights.

Weighted hybrid from Chapter 10; the lecture's 5/20-rating switching rule is not used.

Different ranking signals, shared constraints.

Question	Content-based	Context-aware	Hybrid
Primary signal	Attribute similarity	Rules + completion	Weighted scores
Works without ratings?	Yes	Yes; default history	Yes; base blend
Uses liked exercises?	Yes, when available	No rating-based term	Via content component
Internal components	Cosine similarity	Knowledge + history	Content + P + K + X
Equipment / time rules	Shared	Shared	Shared

engine.py — implementation comparison, not a measured quality ranking

Show the selected method changing the workout.

- 01 Save a profile** Set goal, experience, equipment and time; click Build my workout.
- 02 Select a method** Use the three visible Recommendation method choices in the builder.
- 03 Compare the three** Switch between Content-based, Context-aware and Hybrid.
- 04 Inspect the evidence** Read How to perform, photos and Why this exercise? details.
- 05 Record feedback** Use Log this exercise, then save feedback in Activity.

What has been checked—and what remains open.

28

application tests passed

36

catalogue scenarios checked

0

equipment / budget violations

Supported by checks

Three scoring methods, time budgets, migration and image integrity.

UI checks cover method changes, draft filters, navigation and logging.

Still to evaluate

Recall@10 and NDCG@10 require eligible genuine held-out histories.

No measured best-method result or fitness-outcome validation.

Current Streamlit suite: 28 tests passed; recorded catalogue checks: reports/evaluation.json

Lecture foundations. A gym-specific application.

Content-based

Applies Chapter 4 similarity to exercise attributes.

Context-aware

Adapts Chapter 8 context to workout feasibility and history.

Weighted hybrid

Uses Chapter 10 score blending with project-specific weights.

Next: collect genuine feedback, then evaluate and tune the weights.